Lecture 10: Exponential function.

Question: Is there function $f: \mathbb{R} \to \mathbb{R}$ s.t.

$$\boxed{f' = f} \quad (f(0) = 1).$$

Let's try if out:

$$f_0 = 1 \qquad f_0' = 0.$$

$$f_1 = x + 1 \qquad f_1' = 1$$

$$f_2 = \frac{x^2}{2} + x + 1 \qquad f_2' = x + 1$$

$$f_n = \underbrace{\frac{x^n}{n!} + \frac{x^{n-1}}{(n-1)!} + \dots + 1} \qquad \underline{f_n' = f_{n-1}}.$$

Are we getting anywhere with this?

Yes, kind of: For $x_0 \in \mathbb{R}$

$$f_{n+1}(x_0) - f_n(x_0) = \frac{x_0^{n+1}}{(n+1)!} \longrightarrow 0$$

as $n \to \infty$.

$f_n'(x_0)$

$f_{n+1}(x_0)$

→ The f_n's almost satisfy the eq. $\boxed{f' = f.}$ (*)

We want to define a limit of the functions f_n, because then we may hope that this limit satisfies (*).

<u>In other words:</u> We want a function

$$f(x) \overset{"}{=} \sum_{k=0}^{\infty} \frac{x^k}{k!} "$$

(In an ideal world we even have

$$f'(x) = \sum_{k=0}^{\infty} \left(\frac{x^k}{k!}\right)' = \sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} = f.$$

How can we make this precise?

1) If we can show that $\forall x \in \mathbb{R}$

the seq. $f_n(x) = \sum^{n} \frac{x^k}{k!}$ converges to some numbers $\left(\sum_{k=0}^{\infty} \frac{x^k}{k!} \right)$

then we can define a function

$$f(x) = \left(\lim_{n \to \infty} f_n(x) \right) = \sum_{k=0}^{\infty} \frac{x^k}{k!} .$$

- We say f is the <u>pointwise limit</u> of the functions f_n.

(we also say $f_n \xrightarrow{pt.} f$).

2) Is it true that

$$\frac{d}{dx} \left(\lim_{n \to \infty} f_n \right) = \lim_{n \to \infty} \left(\frac{d f_n}{dx} \right) ?$$

<u>Lemma 1</u> Let $x \in \mathbb{R}$. Then the series

$$\sum_{k=0}^{\infty} \frac{x^k}{k!} \quad \text{converges.} \quad \left(\text{this means } \sum_{k=0}^{n} \frac{x^k}{k!} \right)$$

converges as $n \to \infty$

(Remember: $\sum_{k=0}^{\infty} q^k$ converges if $0 \leq q < 1$).

<u>General criterion:</u> for a series $\sum_{k=0}^{\infty} a_k$ to <u>converge</u>: $\exists N \; \exists r \in [0, 1)$ s.t.

$$\forall n \geq N \qquad |a_{n+1}| \leq r \cdot |a_n|.$$

<u>In our case:</u>

$$a_{n+1} = \frac{x^{n+1}}{(n+1)!} = \frac{x}{(n+1)} \cdot a_n.$$

$$a_n = \frac{x^n}{n!}$$

If $n >> 0$ then $\frac{x}{n+1} < 1$.

Pick N s.t. $\frac{x}{N+1} < 1$ then the ratio criterion is satisfied with $r = \frac{x}{N+1}$.

<u>Defn</u>. We define the function $\exp: \mathbb{R} \to \mathbb{R}$ as

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} \left(= \lim_{n\to\infty} f_n(x) \right);$$

What about the diff. of exp?

Assume we have a seq. of fctn $f_n: \mathbb{R} \to \mathbb{R}$

s.t. $(*)$
- $\exists f: \mathbb{R} \to \mathbb{R}$ st. $\forall x$: $f(x) = \lim f_n(x)$,
- $\forall n$ f_n is diff'.

$\bullet\ \exists g : \mathbb{R} \to \mathbb{R}$ st $g(x) = \lim f_n'(x)$ $\forall x \in \mathbb{R}$.

We may try to prove that

f is diff at some $x \in \mathbb{R}$.

For given $\varepsilon > 0$ we to find $\delta > 0$

st. $\forall x' \in (x-\delta, x+\delta)$.

$$\left| \frac{f(x') - f(x)}{x'-x} - g(x) \right| < \varepsilon$$

$$\left| \frac{f(x')-f(x)}{x'-x} - \frac{f_n(x')-f_n(x)}{x'-x} + \underbrace{\frac{f_n(x')-f_n(x)}{x'-x}}_{f_n'(x_0)} - g(x) \right|$$

<u>Goal</u>: to control

1) $\left| \frac{f(x') - f(x)}{x' - x} - \frac{f_n(x') - f_n(x)}{x' - x} \right|$

2) $\left| f_n'(x_0) - g(x) \right|$

$\forall x', x_0 \in (x - \delta, x + \delta)$ $(*)_2$.

<u>Problem</u> With our assumptions this is impossible!

We need to control

$|f_n - f|$ & $|f_n' - g|$

$\forall x$ in a given interval!

<u>Def</u> We say a seq of functions f_n converges absolutely to f on $S \subseteq \mathbb{R}$ if : $\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n \geq n$ s.t. $\forall x \in S : | f_n(x) - f(x) | < \varepsilon$.

<u>Theorem</u> : In the situation above

(*). If $f_n' \longrightarrow g$ absolutely on an interval (a, b)

then $\forall x \in (a, b)$

$\lim_{n\to\infty} f_n = f$ is diff'ble at x

and $f'(x) = g(x)$.

Pf. Continue with the computation above. We need to find N and δ st. $\forall x' \in (x-\delta, x+\delta)$

$$(\dagger) \quad \left| \underbrace{\frac{f(x') - f(x)}{x'-x}}_{Q} - \underbrace{\frac{f_n(x') - f_n(x)}{x'-x}}_{Q_n} \right| < \varepsilon$$

We need to show that

$Q_n \to Q$ absolutely.

Trick: If $Q_n \to Q$ conv. pointwise & Q_n is absolutely Cauchy then $Q_n \to Q$ absolutely.

Indeed: $|Q_n(x') - Q_m(x')|$

$$= \left| \frac{f_n(x') - f(x)}{x' - x} - \frac{f_m(x') - f(x)}{x' - x} \right|$$

$$= \left| f_n'(x_1) - f_m'(x_2) \right|.$$

becomes arbitrarily small for n, m large!

2) $| f_n'(x_0) - g(x) |$

$$\left| \underline{f_n'(x_0) - g(x_0)} + \underline{g(x_0) - g(x)} \right|$$

<u>Upshot</u>: Apply this to exp!

We have to show that

$$f_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$$

converges to exp uniformly on any interval of the form $(-r, r)$ for $r \in \mathbb{R}$.

Pf. ISTS that the f_n's are uniformly Cauchy on $(-r, r)$.

For this we only the following observation: $(n > m$

$$|f_n(x) - f_m(x)| = \left|\sum_{k=m+1}^{n} \frac{x^k}{k!}\right|$$

$$\le \underbrace{\sum_{k=m+1}^{n} \frac{r^k}{k!}}$$

gets small if $n, m \gg 0$.

$\Rightarrow$ f_n's are un. uniformly Cauchy

$$\Rightarrow \exp'(x) = \lim_{n\to\infty} f_n'(x)$$

$$= \exp(x).$$

$\forall x.$

Complex numbers

<u>Idea</u>. Introduce a new number i, satisfying $i^2 = -1$.

$$\mathbb{C} = \{ x + iy \mid x, y \in \mathbb{R} \}.$$

<u>Addition</u> & <u>Multiplication</u> of the real numbers extend naturally to $\mathbb{C}$ ($\mathbb{R} \subseteq \mathbb{C}$)

$$x \longmapsto x + i \cdot 0$$

Notation: $z = x + iy \in \mathbb{C}$

- $\mathrm{Re}(z) = x \in \mathbb{R}$
- $\mathrm{Im}(z) = y \in \mathbb{R}$.

Fact 1: $\mathbb{C}$ is a field.

(i.e. every $z \in \mathbb{C} \setminus \{0\}$ has a mult. inverse: $z = x + iy$

$$z \cdot \frac{(x - iy)}{x^2+y^2} = \frac{x^2 - i^2 y^2}{x^2+y^2} = 1.$$

Notation: $x - iy = \bar{z}$ (Complex conj. of z)

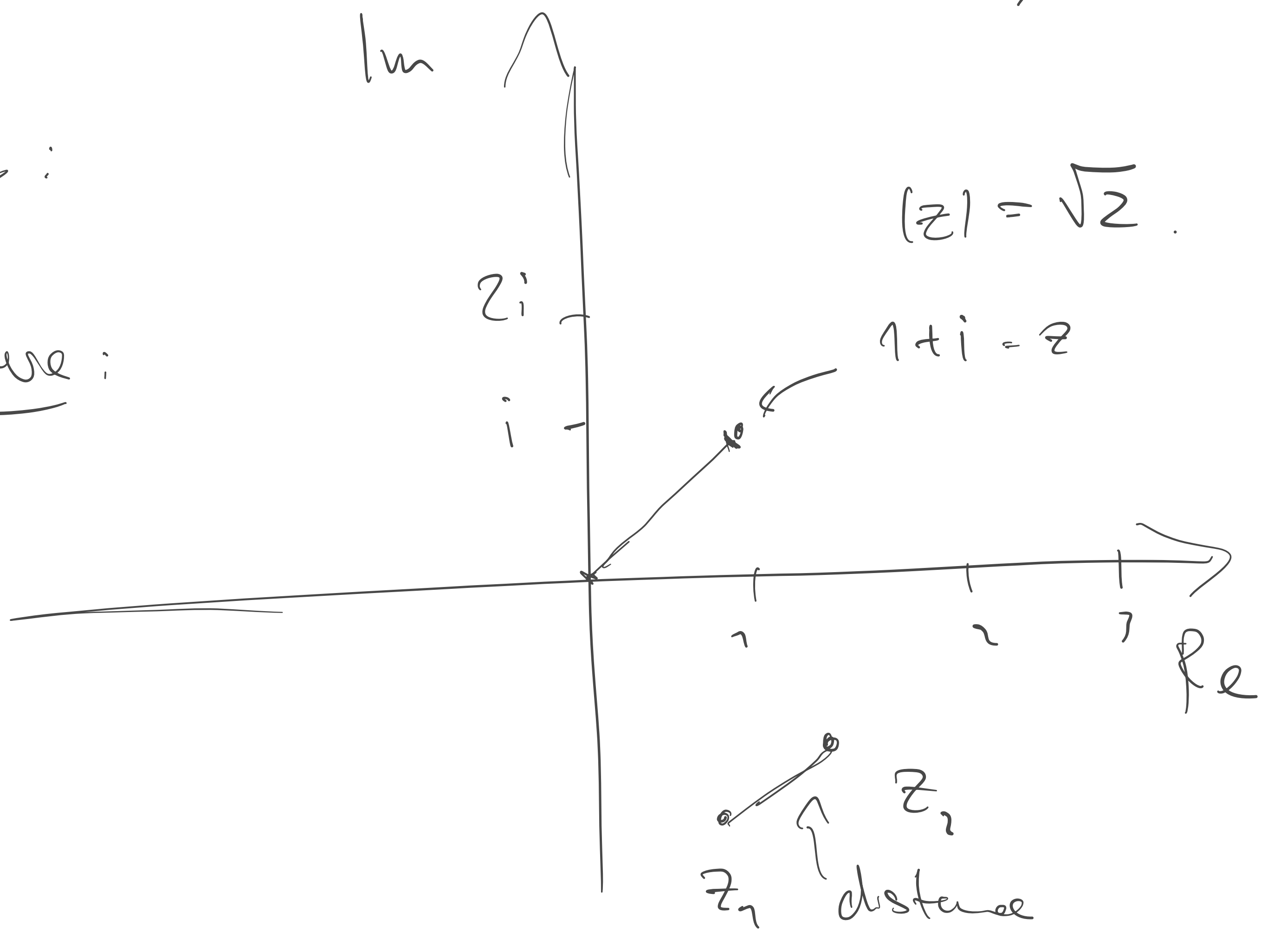
__Fact 2.__ There is a abs. value function $|\cdot|: \mathbb{C} \longrightarrow \mathbb{R}$

- $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
- $|z_1 + z_2| \leq |z_1| + |z_2|$

$$z \longmapsto \sqrt{\underbrace{z \cdot \bar{z}}_{=(x^2+y^2)}}$$

$\mathbb{C}$:

__Picture:__



$|z_1 - z_2|$.

$\rightarrow$ This induces a notion of distance on $\mathbb{C}$

$$(\text{dist}\ z_1 z_2 = |z_1 - z_2|)$$

→ All notions of convergence that we talked about in the course carry over 1-1.

Directions to go into:

- Fundamental theorem of algebra:
  → Every complex polynomial has a root.
- We can work out a theory of complex analysis. In part,

complex derivation.

For example: $f' \text{"="} \lim_{z' \to z} \frac{f(z') - f(z)}{z' - z}$.

→ this leads rich theory of complex analytic functions.

→ 1) General notion of convergence

2) Cauchy-Riemann equations for complex analytic f.

3) Cauchy-Int. formula ...

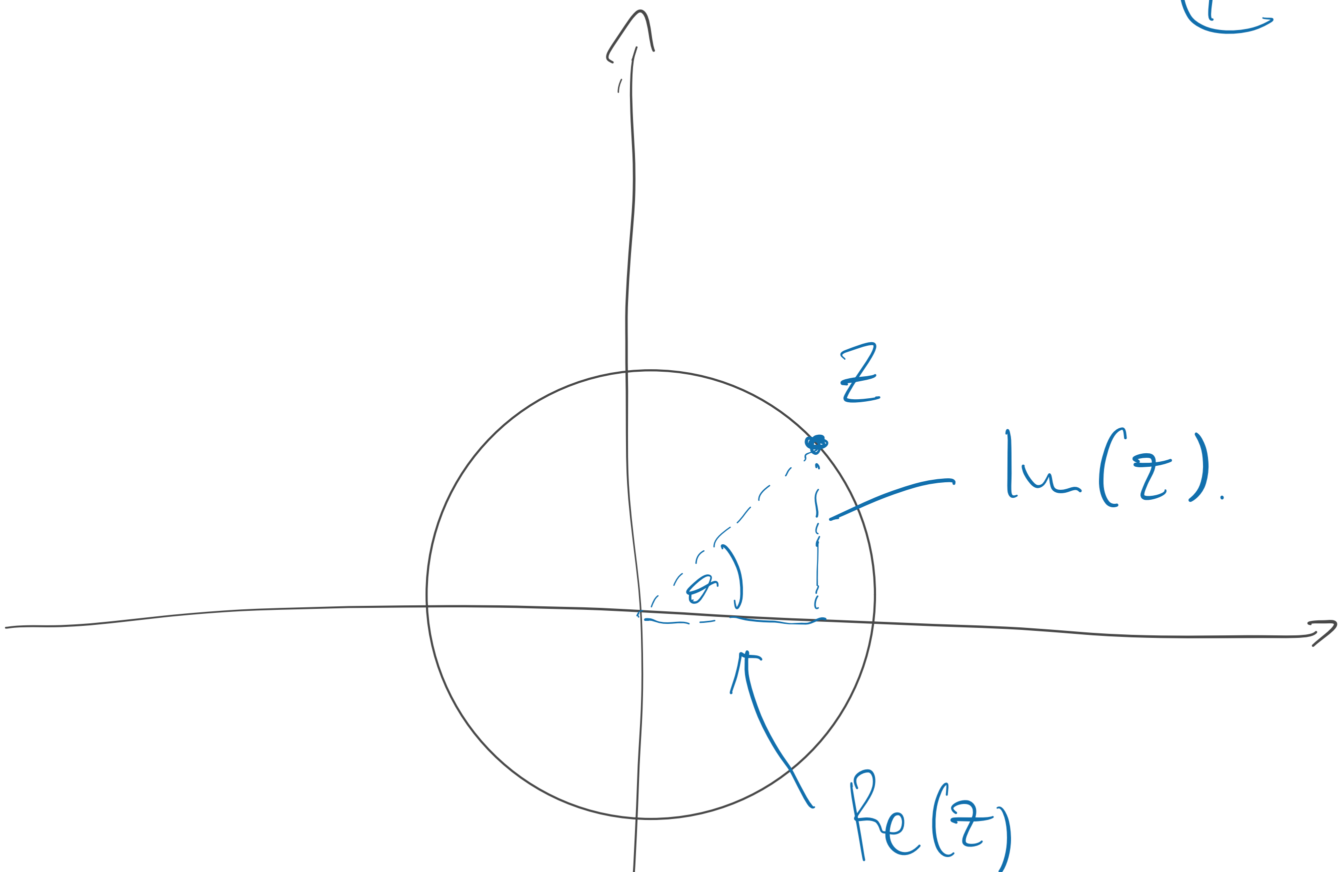
- Power series in complex variables (exp, sine, cosine).

By studying the complex exp function (& sine / cosine) you can prove:

- Euler identity $\exp(i\pi) = -1$.

$$\left[\exp(i\vartheta) = \cos\vartheta + i\sin\vartheta \text{ for } \vartheta \in \mathbb{R}. \right]$$

- Working harder leads to every $z \in \mathbb{C}$ is the form $z = r \cdot \exp(i\vartheta)$ for $r \in \mathbb{R}_{\geq 0}$.



$$z = \mathrm{Re}\, z + i\, \mathrm{Im}(z)$$
$$= \cos\theta + i \sin\theta$$
$$= \exp(i\theta).$$

You can also deduce this purely from the power series defn's of sin & cos.

Another <u>consequence of Euler's identity</u>: You can compute

$$(\cos\theta + i\sin\theta)^n = \exp(i\theta)^n$$

$$= \exp(in\theta)$$

$$= \cos n\theta + i\sin n\theta$$